

CLOUDWATCH

Amazon CloudWatch is a monitoring and observability service in AWS that collects, tracks, and analyzes metrics, logs, and events to help you monitor your applications and infrastructure in real time and take automated actions.

Uses of CloudWatch

- Monitor CPU utilization of EC2 instances
- Track network traffic and disk usage
- Create alarms to trigger Auto Scaling
- Collect and store application logs
- Detect and troubleshoot issues
- Monitor billing and usage

Simple Notification Service (SNS)

Amazon Simple Notification Service (SNS) is a service in Amazon Web Services used to send notifications and messages to multiple subscribers.

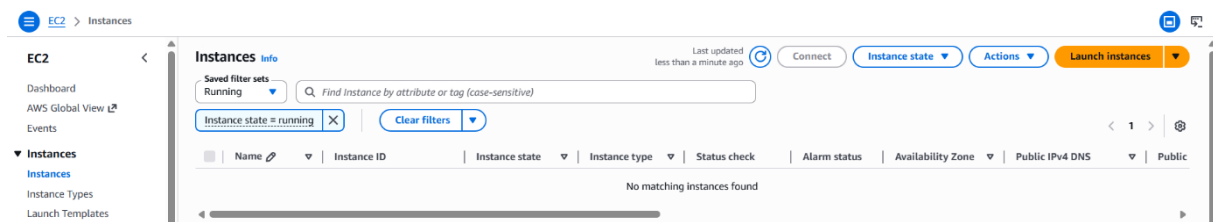
Types of SNS Protocols

- Email → Sends notification to email
- Email-JSON → Sends message in JSON format
- SMS → Sends message to mobile number
- HTTP → Sends message to HTTP endpoint
- HTTPS → Sends message securely to HTTPS endpoint
- SQS → Sends message to queue (Amazon Simple Queue Service)
- Lambda → Triggers AWS Lambda function
- Application → Sends push notification to mobile apps

HANDS-ON FOR CLOUD WATCH AND SENDING ALERT MESSAGE THROUGH EMAIL

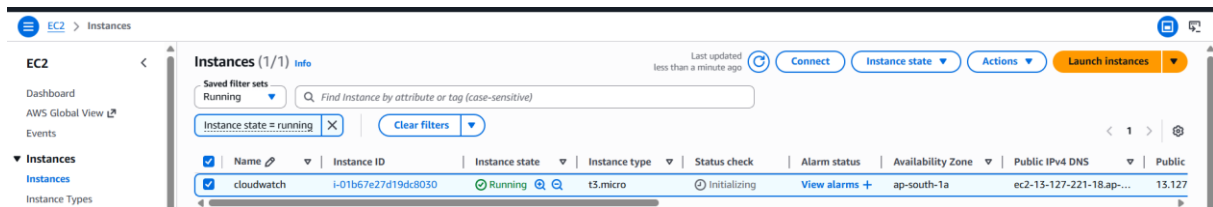
Screenshot: Open EC2

- Go to Amazon Web Services
- Click EC2
- Click Instances



Screenshot: Instance Created

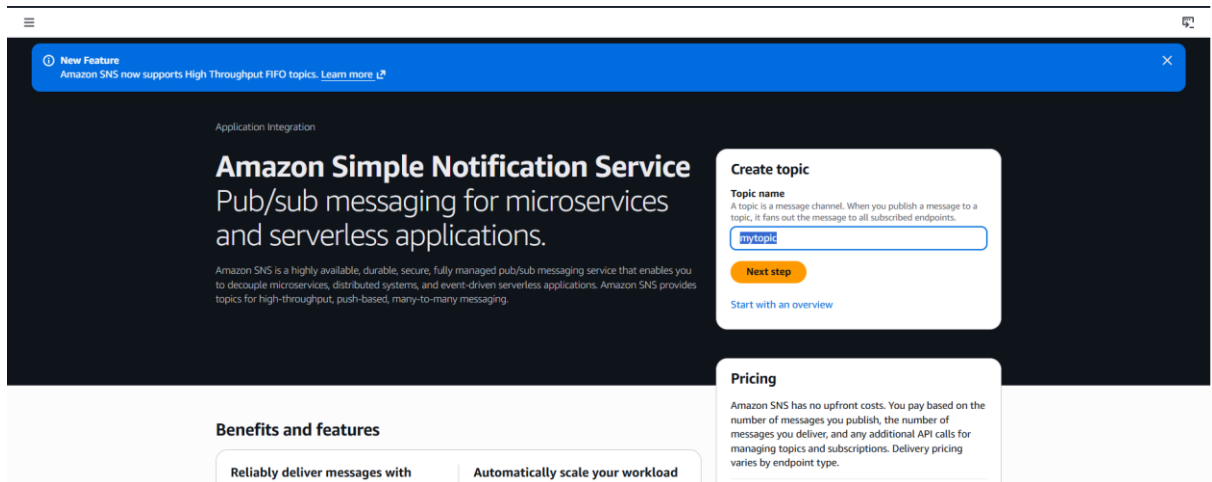
- Amazon Linux instance created
- Instance is running



Creating sns (Simple message service)

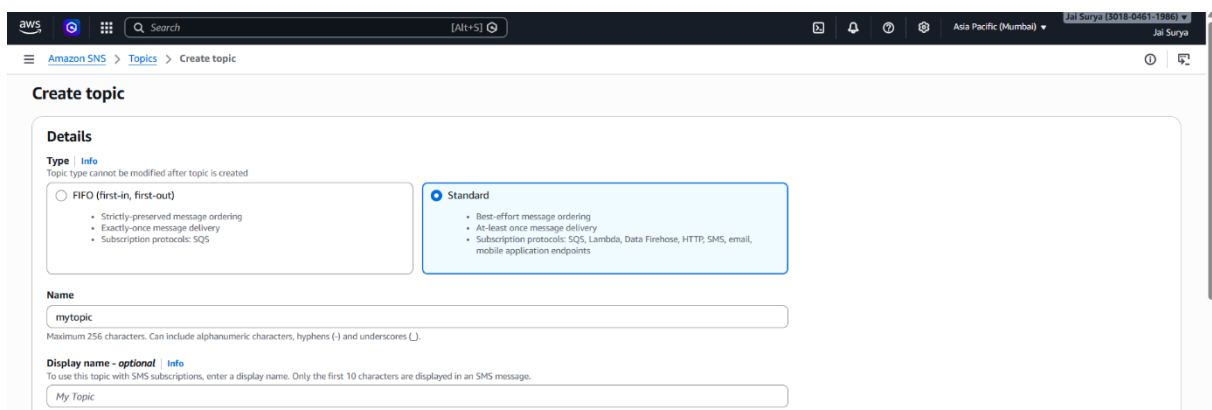
Screenshot: Create SNS Topic

- Go to Amazon Web Services
- Click SNS
- Enter topic name
- Click Next step



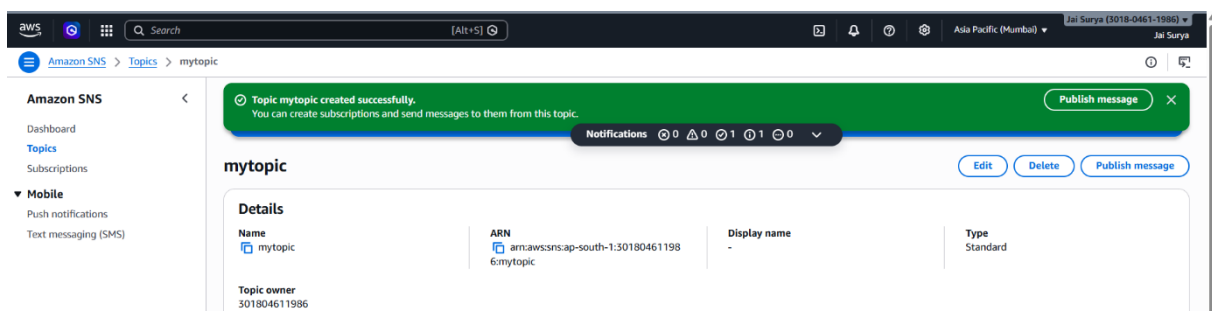
Screenshot: Select Topic Type

- Choose Standard
- Click Create topic



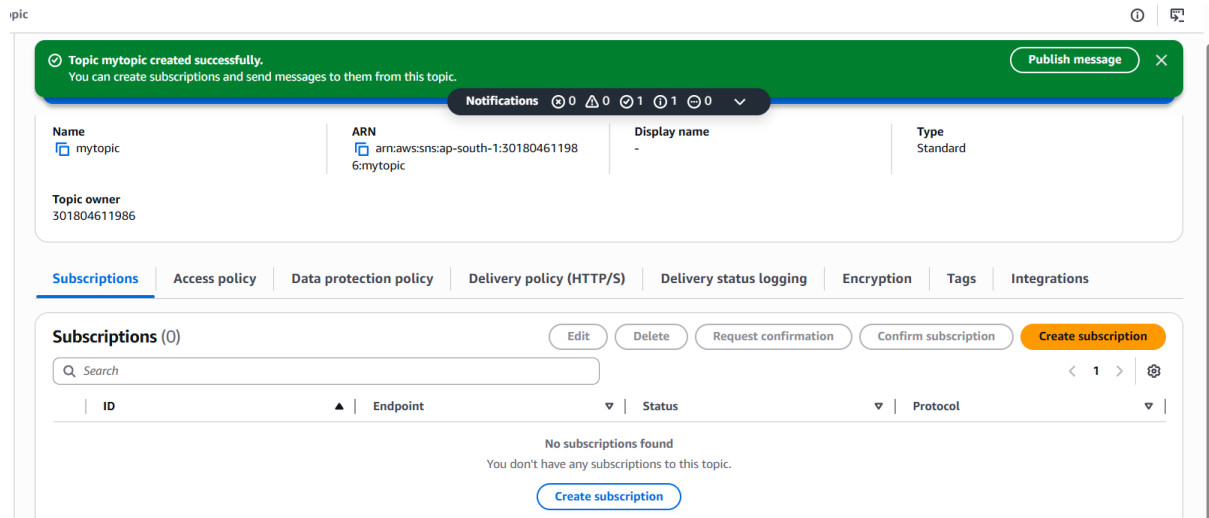
Screenshot: Topic Created

- SNS topic created successfully



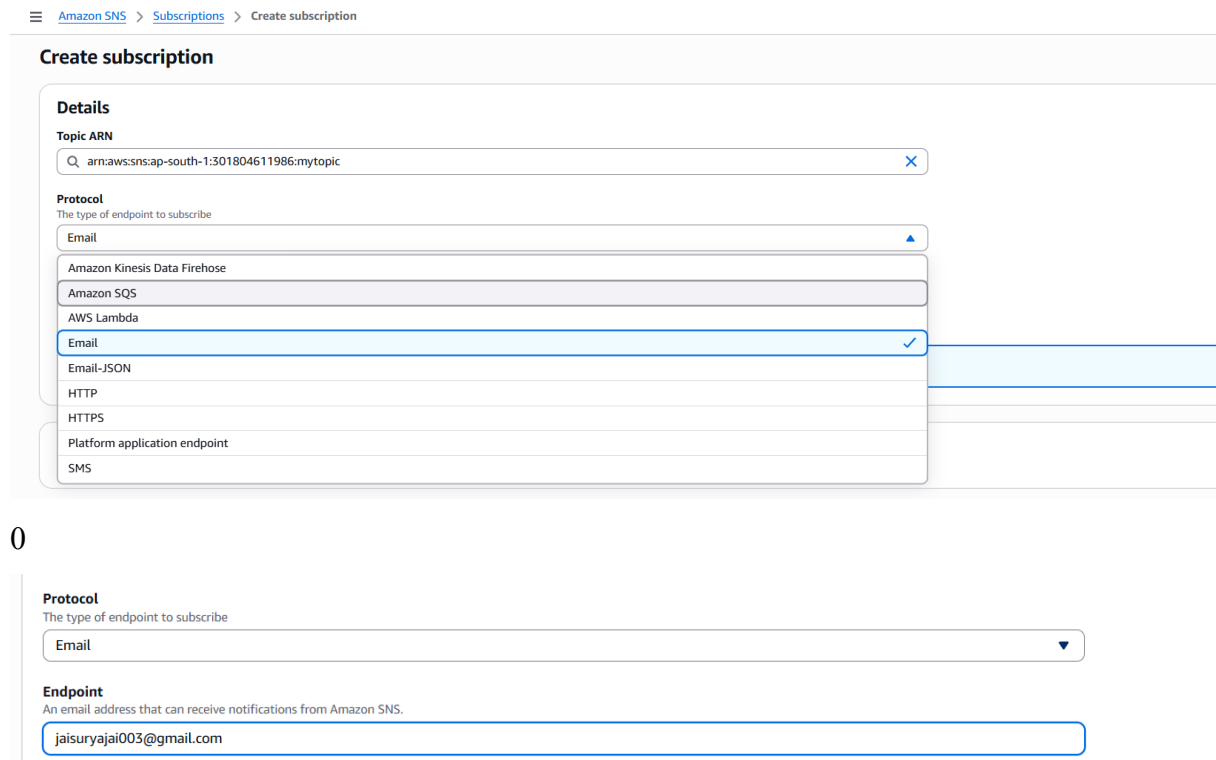
Screenshot: Create Subscription

- Click Create subscription



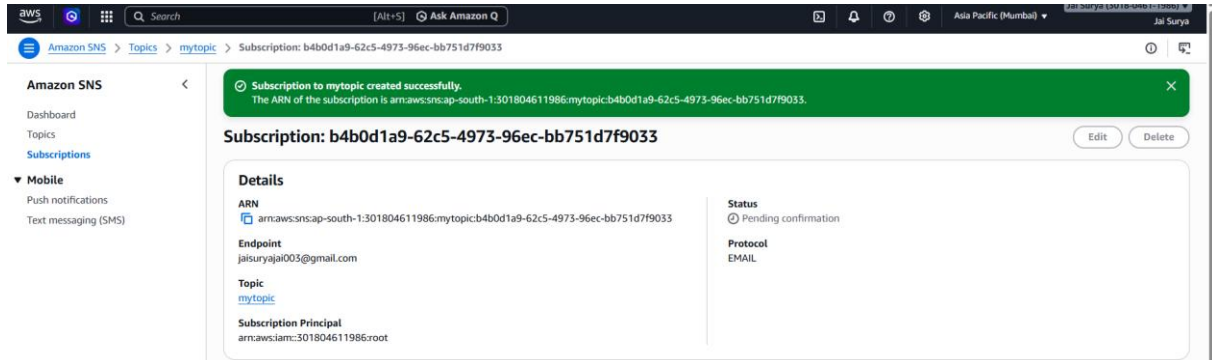
Screenshot: Configure Subscription

- Choose protocol as Email
- Enter email ID
- Click Create subscription



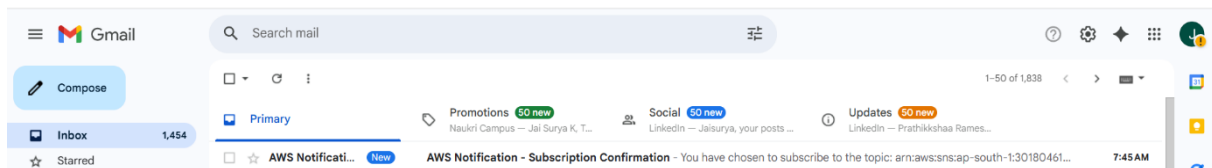
Screenshot: Subscription Pending

- Subscription created successfully
- Status shows pending confirmation



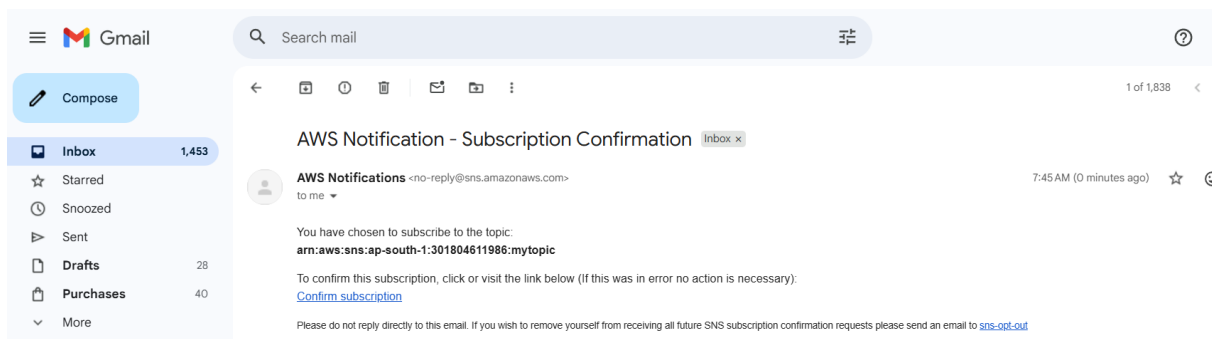
Screenshot: Confirm Subscription

- Go to Gmail
- Open AWS notification email



Screenshot: Subscription Confirmed

- Click Confirm subscription
- Subscription confirmed successfully





Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:ap-south-1:301804611986:mytopic:b4b0d1a9-62c5-4973-96ec-bb751d7f9033

If it was not your intention to subscribe, [click here to unsubscribe](#).

Screenshot: Status Confirmed

- Subscription status shows confirmed

Subscription: b4b0d1a9-62c5-4973-96ec-bb751d7f9033

Details

ARN

 arn:aws:sns:ap-south-1:301804611986:mytopic:b4b0d1a9-62c5-4973-96ec-bb751d7f9033

Endpoint

jaisuryajai003@gmail.com

Topic

[mytopic](#)

Subscription Principal

arn:aws:iam::301804611986:root

Status

 Confirmed

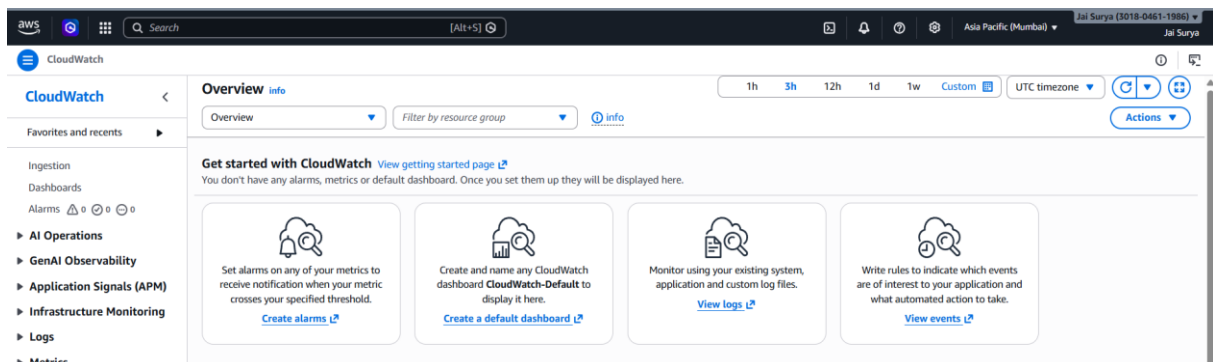
Protocol

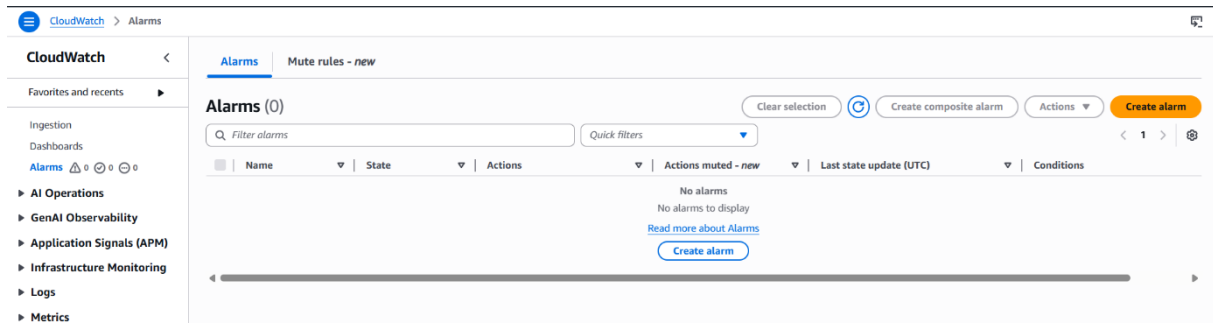
EMAIL

CREATING ALARM

Screenshot: Open CloudWatch Alarms

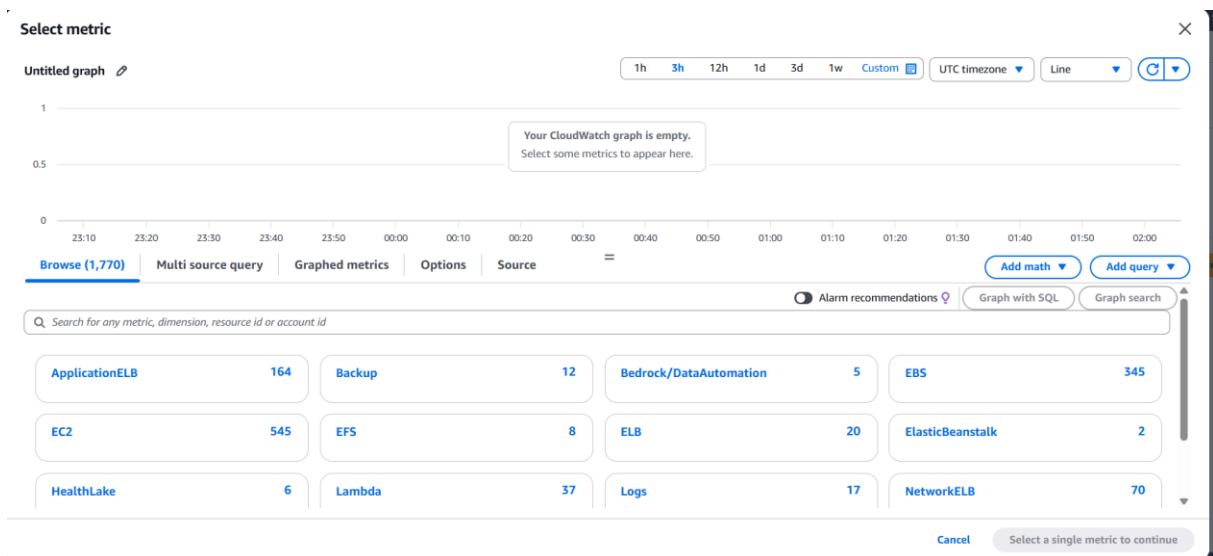
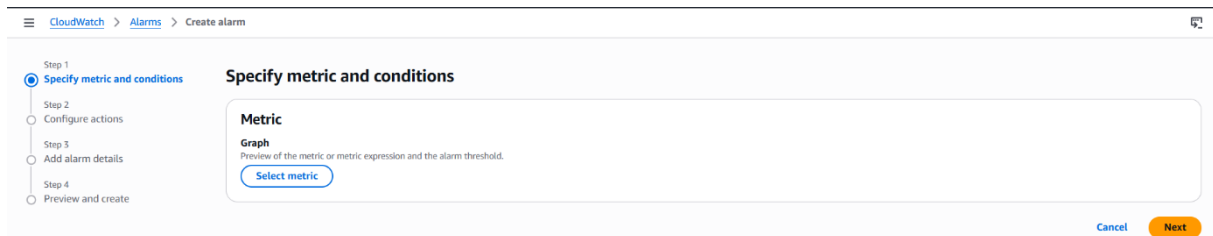
- Go to Amazon Web Services
- Click CloudWatch
- Click Alarms
- Click Create Alarm





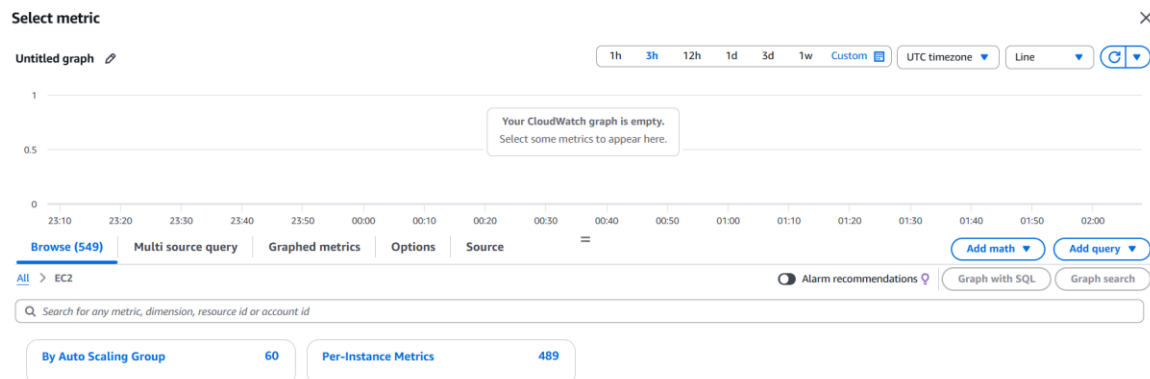
Screenshot: Select Metrics

- Click Select metric
- Click EC2 in metrics



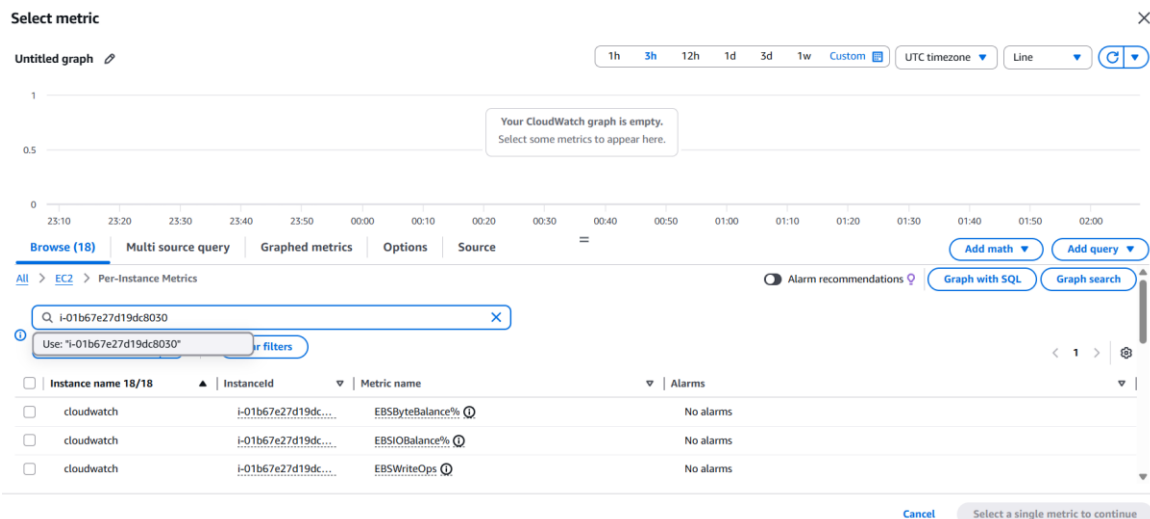
Screenshot: Select Per-Instance Metrics

- Click Per-Instance Metrics



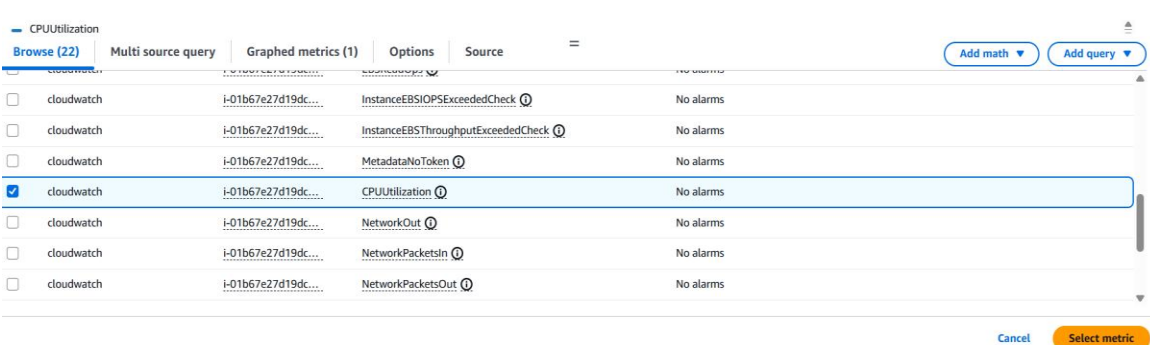
Screenshot: Select Instance Metric

- Paste the instance ID in search box
- Select the instance ID



Screenshot: Select Metric

- Select CPU Utilization
- Click Select metric



Screenshot: Configure Threshold

- Select Static
- Choose Greater than
- Enter value as 1
- Click Next

The screenshot shows the 'Conditions' section of the 'Create alarm' wizard. Under 'Threshold type', the 'Static' option is selected with the subtext 'Use a value as a threshold'. The 'Whenever CPUUtilization is...' section has 'Greater' selected with the subtext '> threshold'. The 'than...' section has a text input field containing '1' with the subtext 'Must be a number.' At the bottom right, there are 'Cancel' and 'Next' buttons.

Screenshot: Configure Notification

- Select existing SNS topic
- Choose the created topic
- Click Next

The screenshot shows the 'Configure actions' section of the 'Create alarm' wizard. Under 'Notification', the 'In alarm' trigger is selected. The 'Send a notification to the following SNS topic' section has 'Select an existing SNS topic' chosen. A search bar shows 'mytopic' as the selected topic. Below, an email endpoint 'jaisuryajai003@gmail.com' is listed. At the bottom, there is an 'Add notification' button.

Screenshot: Alarm details

- Enter alarm name
- Enter description for alarm

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 1: Specify metric and conditions
Step 2: Configure actions
Step 3: Add alarm details
Step 4: Preview and create

Add alarm details

Name and description

Alarm name
High-CPU-Utilization-ASG

Alarm description - optional [View formatting guidelines](#)

Edit Preview

Triggers when CPU utilization exceeds 1% to scale out EC2 instances

Up to 1024 characters (67/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Screenshot: alarm created

- Alarm created successfully
- Status shows ok

CloudWatch > Alarms

CloudWatch < Alarms Mute rules - new

Alarms (1)

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	High-CPU-Utilization-ASG	OK	Actions enabled	-	2026-04-02 02:24:27	CPUUtilization > 1 for 1 datapoints w minutes

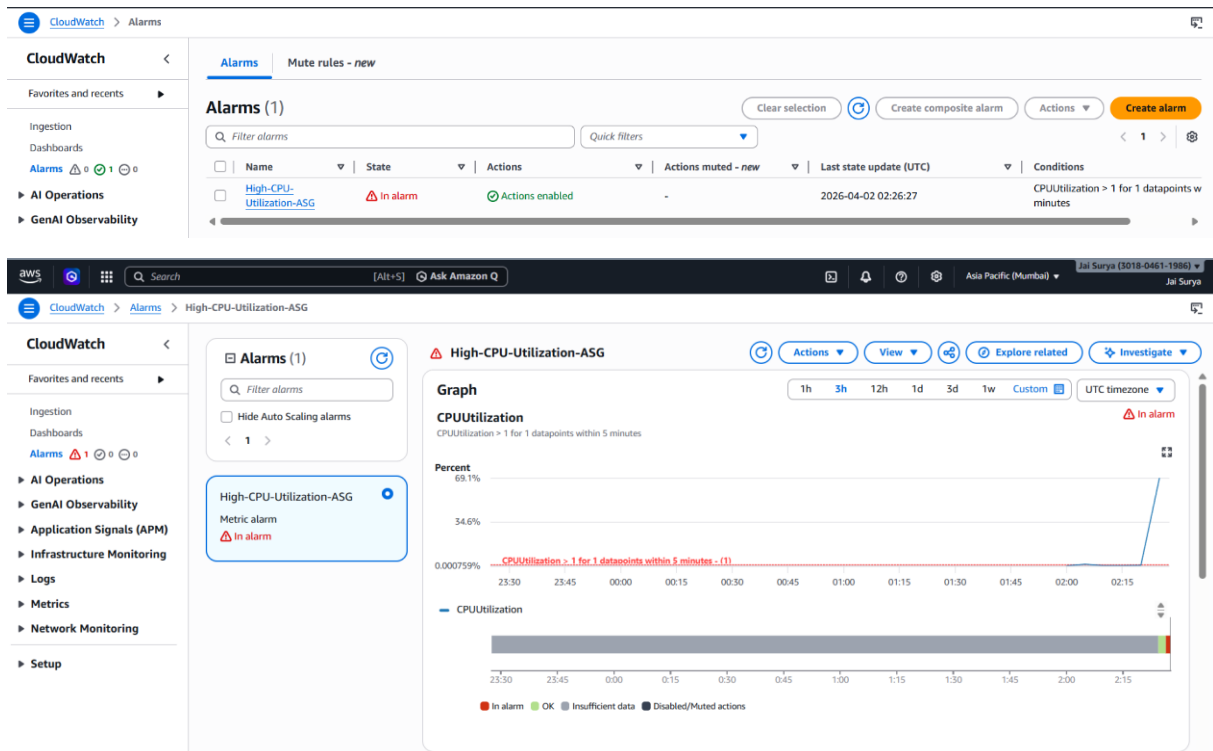
Screenshot: fake cpu utilization

- Connect and linux terminal
- Execute commands to increase cpu utilization
 - sudo su
 - yum install stress -y
 - stress --cpu 4 --timeout 300

```
Complete!
[root@ip-172-31-36-201 ec2-user]# stress --cpu 4 --timeout 300
stress: info: [22174] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
^C
[root@ip-172-31-36-201 ec2-user]# ^C
[root@ip-172-31-36-201 ec2-user]# stress --cpu 4 --timeout 300
stress: info: [26204] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
^C
[root@ip-172-31-36-201 ec2-user]# stress --cpu 4 --timeout 300
stress: info: [26209] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
```

Screenshot: Alarm state in warning

- Alarm state shows warning sign



Screenshot: message sent successfully

- Open the mail app
- Aws alert message received

